Microsoft Elevate

Amazon Sales Analysis

PRESENTED BY

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OUTLINE
AMAZON SALES ANALYSIS

Data Source

Table: Amazon_sales with a supporting DateTable

Key Fields:

product_category, customer_region, payment_method

total_revenue, profit, quantity_sold, discount_percent

rating, review_count

Date dimensions: Year, Quarter, Month

PROBLEM STATEMENT:

Amazon, as a large-scale e-commerce platform, generates vast volumes of transactional data across multiple product categories, regions, and customer segments. However, without structured analysis, this data remains underutilized — making it difficult for business stakeholders to identify performance gaps, optimize pricing strategies, or allocate resources effectively.

PROPOSED SOLUTION:

The proposed system aims to address the challenge of analyzing and visualizing Amazon sales data to ensure accurate tracking of revenue, profit, regional performance, and product demand. This involves leveraging **data analytics and interactive visualization techniques** to uncover actionable business insights. The solution consists of the following components:

Data Collection

- Gather historical Amazon sales transaction data including product category, customer region, payment method, quantity sold, revenue, profit, discount percentage, and customer ratings from Kaggle.
- Utilize supporting data sources such as a structured **Date Table** to enable time-based analysis across Year, Quarter, and Month dimensions.
- Incorporate product-level metadata including review counts and ratings to support performance benchmarking.

Data Preprocessing

- Clean and preprocess the collected data to handle missing values, outliers, and data type inconsistencies within Power Query Editor.
- Apply **feature engineering** to derive calculated columns and DAX measures such as Total Revenue, Total Profit, Average Rating, and Top N Product rankings.
- Establish relationships between the sales table and date table to enable accurate time intelligence and cross-filtering across all report pages.

Deployment

- Develop a **user-friendly, multi-page Power BI Dashboard** that provides real-time interactive insights into sales trends, regional performance, discount impact, and product rankings.
- Deploy the solution on **Power BI Service** — a scalable and reliable cloud platform — ensuring accessibility across devices with role-based sharing and scheduled data refresh.
- Incorporate **slicers, navigation buttons, and cross-filtering** to allow non-technical stakeholders to explore data independently without requiring SQL or coding knowledge.

SYSTEM APPROACH:

The System Approach outlines the overall strategy and methodology for developing and implementing the Amazon Sales Analysis dashboard.

System Requirements

- Microsoft Power BI Desktop is required to build and run this project, and a stable internet connection for publishing to Power BI Service.

Libraries & Features Required to Build the Model

- Power BI's built-in suite of tools is utilized to develop this solution, including **Power Query Editor** for data cleaning and transformation, **Filters** for create a table of top-3 products by Average Rating and Quantity Sold , **Data Modeling** for establishing relationships between the Amazon_sales and DateTable, and core visuals such as **Map**, **Donut Chart**, **Bar Chart**, **Scatter Chart**, **Line Chart**, **Card Visual**, **Slicers**, **and Navigation Buttons** to build the complete 4-page interactive dashboard.

RESULT:

The Amazon Sales Analysis Power BI dashboard successfully delivered a comprehensive, interactive, and visually appealing 4-page report that transforms raw Amazon sales data into meaningful business insights. The **Overview page** provides a high-level summary of total revenue, profit trends over time, and category-wise performance through cards, line charts, and column charts. The **Regional Analysis page** identifies geographic revenue distribution across customer regions using map visuals, donut charts, and bar charts. The **Payment & Discount page** uncovers the relationship between discount percentages and profitability, along with payment method preferences through scatter charts and donut charts. Finally, the **Product Performance page** benchmarks top products by average rating and quantity sold using detailed tables and line charts — collectively enabling stakeholders to make data-driven decisions on sales strategy, regional targeting, pricing optimization, and product prioritization.

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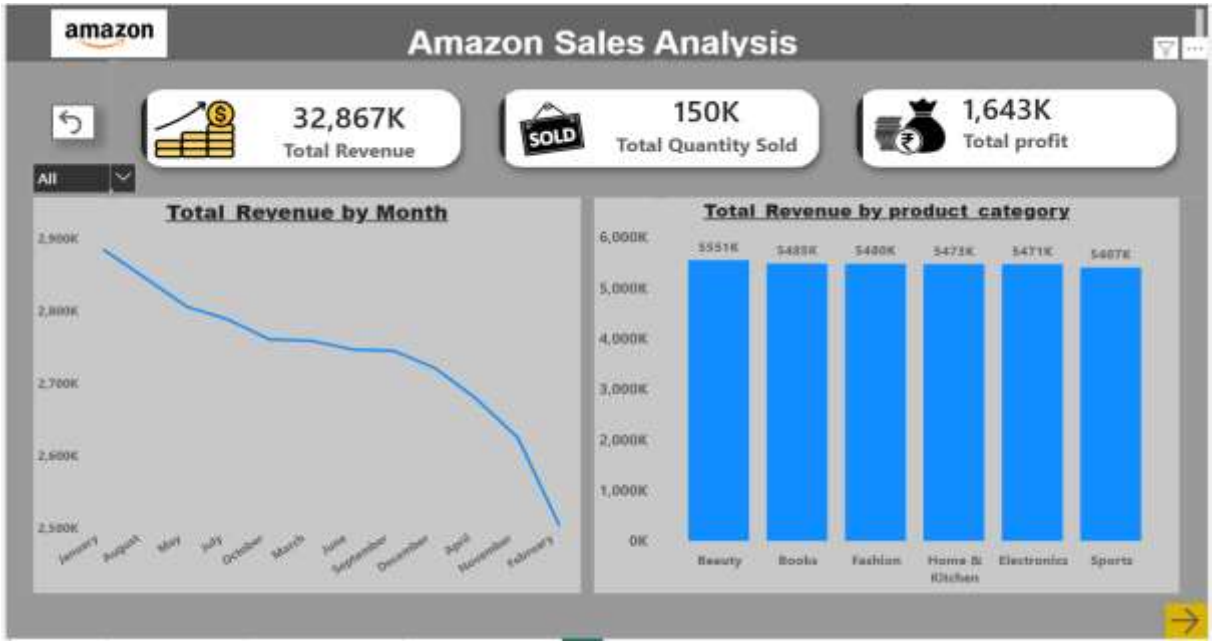

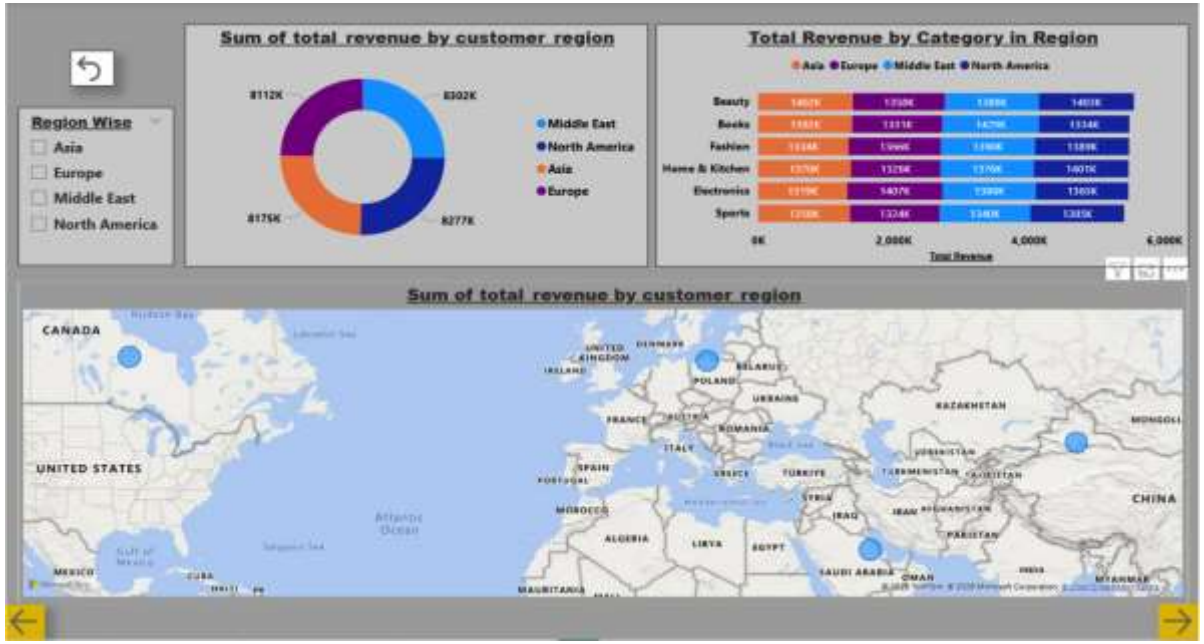

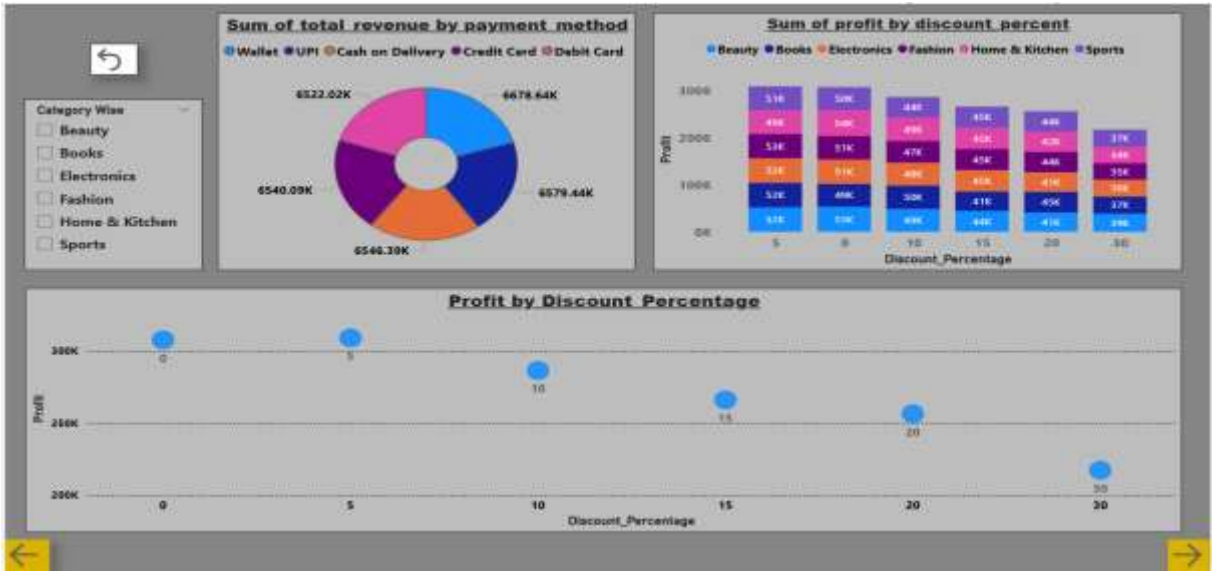

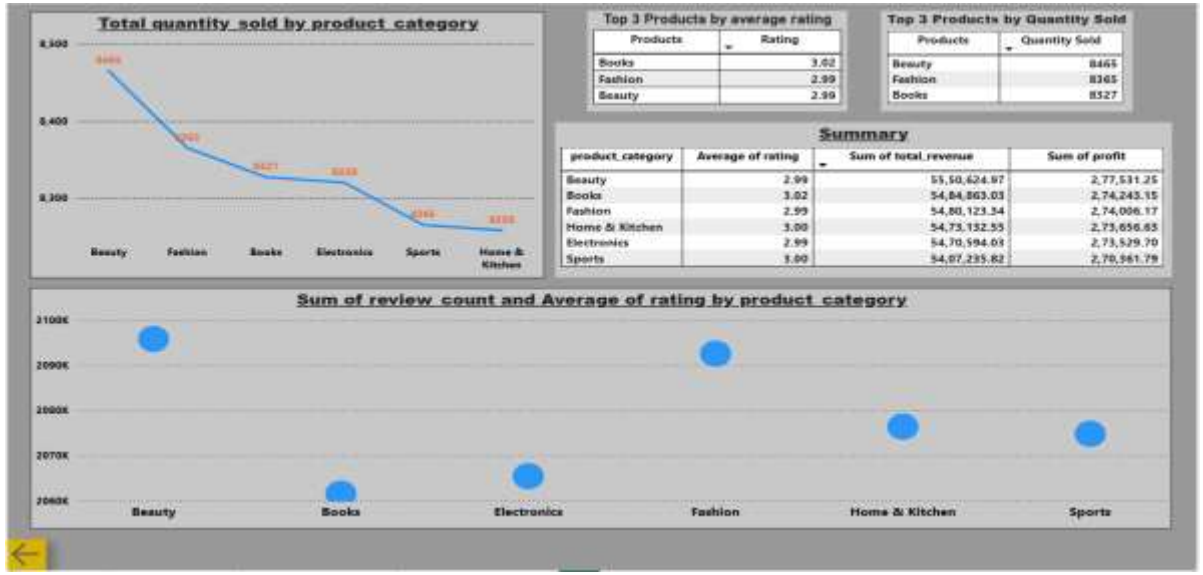

Top 3 Products by average rating

Products	Rating
Books	3.02
Fashion	2.99
Beauty	2.99

Top 3 Products by Quantity Sold

Products	Quantity Sold
Beauty	8465
Fashion	8365
Books	8327

Summary

product_category	Average of rating	Sum of total_revenue	Sum of profit
Beauty	2.99	55,50,624.97	2,77,531.25
Books	3.02	54,84,863.03	2,74,243.15
Fashion	2.99	54,80,123.34	2,74,006.17
Home & Kitchen	3.00	54,73,132.55	2,73,656.63
Electronics	2.99	54,70,594.03	2,73,529.70
Sports	3.00	54,07,235.82	2,70,361.79

CONCLUSION:

The Amazon Sales Analysis Power BI dashboard successfully transforms raw Amazon sales data into clear, actionable insights across four key dimensions — revenue trends, regional performance, payment behavior, and product benchmarking. By leveraging Power Query, DAX measures, and interactive visuals, the solution empowers stakeholders to monitor business health, identify high-performing regions and products, and optimize discount strategies in real time. The dashboard serves as a centralized decision-making platform that simplifies complex data, enabling business teams to drive revenue growth and improve overall operational efficiency effectively.

FUTURE SCOPE:

The Amazon Sales Analysis dashboard can be further enhanced by integrating **machine learning models** to predict future sales trends and product demand. Incorporating **customer demographic data** and real-time data streaming via **Power BI Service** would enable deeper segmentation and up-to-date insights. Adding **competitor pricing analysis** and **seasonal trend integration** would make the solution a truly comprehensive, future-ready business intelligence platform.

REFERENCES:

Kaggle Dataset Link: https://www.kaggle.com/datasets/litonislam/amazon-sales-analysis-cleaned-data/data

GitHub Link: https://github.com/samenushyam65-eng/Amazon-Sales-Analysis-.git

Drive Link: https://drive.google.com/drive/folders/1NrHoPlRf9RKj1BgCA80F2NWfED-XXVCj?usp=drive_link

Thank You